Classroom notes

The Bottomless Popcorn Paradox · for teachers, librarians and reading-group leaders

At a glance

Grades	1–3 (ages 6–9). Works as a read-aloud from Grade 1; strongest for independent reading at ages 7–9.
Topic	Salt, thirst and water — how a body asks for what it needs — and the habits of a scientist: noticing, checking a reason, fair tests, data.
Story	Eight short chapters, about 1,700 words — roughly 20 minutes read aloud.
Back matter	Lumi's Secret Lab Notes (pp. 38–49) · Think Like a Scientist (pp. 50–51) · Lumi's Word Lab, 22 words (pp. 53–65) · Jiwoo's "Prove It" Tracker (pp. 66–71) · For the Grown-Up Lab Partner (pp. 72–74)

Big ideas

- Bodies send signals — a dry mouth, a tight head — and a signal can be noticed or missed.
- Blood keeps a just-right mix of salt and water. Thirst is the body asking for water to balance it.
- Noticing is a skill. Some bells are loud, some are quiet, and every scientist can practice listening.
- The first reason you notice is not always the right one. Scientists check a reason before they believe it.
- A blank day and a question mark are real data.

How this book handles food — please keep it this way in class

- No food is called good or bad, and there are no sodium or calorie counts. Please keep it that way in class.
- No one's body is discussed. Keep comments on feelings and signals, never on size or weight.
- Don't collect, display or grade students' food records. The tracker belongs to the child and stays at home.
- Plan activities without food or drink, and don't use snacks as rewards during the unit.
- Praise a blank day and a question mark. Let a wrong reason stand for a minute; ask one question instead of correcting it.

Sensitivity notes

Salt and fluid plans. Some students follow a salt or fluid plan from their doctor — less salt for some kidney or heart conditions, extra salt for others, such as cystic fibrosis. Follow their care plan and their family's lead, and never ask them to explain it to the class.

Water and bathroom breaks. This story turns on thirst and a bathroom emergency. Some students need water or the bathroom more often for medical reasons, and some feel embarrassed about it. Make both easy to get, keep the talk on Jiwoo, and never comment on a particular child's trips.

Headaches. Some students live with headaches or migraines that have nothing to do with salt or water. Treat Jiwoo's headache as one kid's case — the book "never promises a headache" (p. 74) — and point families to the doctor box on the Trackers page.

Food at home. Ramen, pizza, chips and pretzels — foods from Jiwoo's week and the rules page — are everyday food in many homes and cultures, and not every child has snacks to track. Nothing in the book calls a food bad, and a blank day is a real result. Keep every activity food-free.

Different bells. Some children notice body signals early and some late, for many reasons. Use the book's own line, "Neither bell is broken," and never point out a particular child's bell.

Body talk. If a student comments on someone's body, redirect to what bodies do: "In this book we're scientists of how bodies feel and work."

Three short lessons

Lesson 1 · Read & Notice *40 min*

5 min	Cover prediction: what could bottomless mean? Record predictions on the board.
15 min	Read aloud. Stop after Chapter 2 (predict), Chapter 5 (what is Jiwoo's body doing?) and Chapter 8 (Jiwoo's two lessons).
10 min	Talk: choose two or three discussion questions.
7 min	Introduce Think Like a Scientist (p. 50): Notice · Wonder · Test · Report.
3 min	Exit ticket: one thing I noticed, one thing I wonder.

Lesson 2 · Model It *40 min*

10 min	Re-read Lab Notes 1–4 (pp. 41–44): the gap, salt pulls water, the Thirst Department, one attention.
20 min	Activity: the Thirst Department Game (active) or Draw the Gap (seated).
10 min	Debrief: explain the bell in your own words. Vocabulary: balance, dilute, thirst, hypothalamus, attention.

Lesson 3 · Check a Reason *40 min*

15 min	Activity: The Paradox Sort (card sort).
15 min	Data detectives: analyse Jiwoo's filled-in week (pp. 69–70) with the questions below.
10 min	Write a fair test on paper — or check Jiwoo's new reason with Lumi's four questions (p. 47). Optional: send home the blank tracker with the family letter.

Data detectives — Jiwoo's week (pages 69–70)

Jiwoo's week (pages 69–70) is made-up data, which makes it safe to analyse together — no student shares their own food. Ask the questions in order; teacher notes on the right.

Question	Teacher note
On how many days did Jiwoo write something down?	Four — Days 1, 4, 5 and 6. Days 2, 3 and 7 are blank, on purpose (p. 73): a blank day is a real result.
On which two days did she drink water zero times? What did she notice each time?	Day 1, the popcorn bucket: dry mouth and a tight-hat headache — noticed only when Maya asked. Day 5, pretzels and string cheese (she forgot): mouth a little dry by dinner.
On Days 4 and 6 she drank water while she ate, and felt fine. Does that prove water keeps the dry mouth and headache away?	Not yet — one kid, a few days. Look closer: Day 4 came after soccer practice, and on Day 6 Maya passed the pitcher — Jiwoo didn't ask for it. It's a pattern worth testing, one change at a time.
Day 5 is an open case: thirst, or "just… Wednesday"? Why is writing down both smart?	A scientist records every possible explanation, then checks one. ("Just Wednesday" is a joke — ask students how she could rule it out.)
In Jiwoo's own words (p. 70), what made the popcorn day different?	"Salty AND ninety minutes AND a movie so good I forgot I had a body." Her tracker on p. 69 adds a fourth: water 0 times in 90 minutes.

Jiwoo ends her week with two maybes. What are they — and what would she need to do to find out which is true?	"Maybe the bell was always ringing at the same volume. Maybe I'm just getting better at picking up the phone." More weeks, the same notes, one change at a time. Lumi's note on p. 48 says listening can be practiced.

Science background for teachers

Blood keeps a steady mix. Sodium is the main salt in the fluid around our cells, and the body keeps the blood's saltiness within a narrow range. Special cells in the hypothalamus sense when blood gets saltier; the brain then switches on thirst and releases a hormone, ADH, that tells the kidneys to save water.

A salty snack tips the mix a little. In a study of adults, one salty meal nudged blood sodium up slightly. Water moves by osmosis out of cells toward the saltier fluid around them, so cells lose a little water: far less than the drawing on page 42 shows. Blood is never salt-free, which is why Lumi's graph on page 41 is best read as "salt I ate."

Thirst is the bell; a headache is a different clue. Thirst is the urge to drink, often with a dry mouth. Going without water can bring on a headache in some people, but researchers say the mechanism isn't fully understood, and the international headache classification has no separate "dehydration headache." Headaches have many causes, which is why the corrections keep the two apart.

The bell rings early. A 2002 review about adults concluded that thirst starts before the body is short enough of water to count as dehydrated. Experts draw that line in different places, and the review left out hard exercise and hot weather. People differ in how readily they feel thirsty; we found no studies of this in children.

Attention turns body signals down, not off. What we notice depends on where our attention is. People who listened to outside sounds while exercising reported less tiredness, and distraction moderately reduces children's pain during shots and other needle procedures. Sensing the inside of your body is called interoception; in adults, months of practice brought small improvements.

One cause, or several? Kidneys make urine at a fairly steady rate, and the bladder stores it until it is emptied. Jiwoo's bathroom emergency was probably a mix of the water after soccer, a long film and skipping the bathroom beforehand (page 8). In rested adults, most of the extra urine from a big drink comes within a couple of hours, so timing matters too. "What else was true that day?" is the right question.

Standards connections

Short descriptions in our own words — check the official documents for exact wording. An honest note first: in U.S. science standards, the body's inner organs first appear in Grade 4 and energy from food in Grade 5. At Grades 1–3, this book supports the practices of science rather than a body or nutrition standard.

Science practices (Grades K–2 and 3–5)

Practice	What it looks like	Where it shows up
Asking questions	Ask questions you can test — in Grades 3–5, ask what would happen if one thing changed.	The Paradox Sort · My Case Report
Using models	Use a model to show how something works — and notice how the model differs from the real thing.	Thirst Department Game · Draw the Gap
Fair tests	Plan a fair test that changes one thing at a time.	The Paradox Sort · My Case Report
Working with data	Record observations, look for patterns and compare them with predictions.	Data detectives · trackers
Explaining with evidence	Build an explanation from what you observed.	Discussion · Sign Detective
Cause and effect	Things that happen together are not always cause and effect.	The Paradox Sort · Discussion
Systems	The parts of a system work together.	Thirst Department Game

Common Core State Standards — English Language Arts

Code	In short	Where it shows up
RL.1.3 · RL.2.3 · RL.3.3	Describe characters, how they respond to events, and how their actions move the story.	Read-aloud and discussion
RL.2.7 · RL.3.7	Use the pictures to understand characters, setting and plot — and what the art adds to the words.	Read-aloud picture walk
RI.2.3 · RI.3.3	Describe how scientific ideas connect, using sequence and cause-and-effect language.	Lab Notes · Draw the Gap · Thirst Department Game
RI.2.4 · RI.3.4 · L.2.4 · L.3.4	Work out the meaning of subject words and phrases.	Word Lab Workout
SL.1.1 · SL.2.1 · SL.3.1	Take part in conversations about texts with partners and groups.	Discussion · Paradox Sort · Sign Detective

Codes refer to the Common Core State Standards for English Language Arts. © Copyright 2010. National Governors Association Center for Best Practices and Council of Chief State School Officers. All rights reserved.

Health education skills

Skill	What it looks like	Where it shows up
Influences	Notice what influences health choices — including media and labels.	Sign Detective
Checking claims	Ask whether health information and product claims can be trusted.	Sign Detective · Discussion
Asking for help	Say how you feel, and know when to tell a trusted adult.	Discussion · trackers · Lumi Holds Up a Flag
Decisions	Compare options, then reflect on how a choice turned out.	Discussion · Data detectives

IB Primary Years Programme

Theme	Why it fits	Where it shows up
Who we are	An inquiry into the nature of the self, including physical and mental health.	The whole unit

Vocabulary — Lumi's Word Lab

Words for every case: case study · subject · data · observe · hypothesis · experiment

Words from this case: paradox · sodium · salt · balance · dilute · thirst · hypothalamus · body signal · dehydration · hydration · symptom · headache · attention · interoception · misattribution · cause

When a student says…

"So salty food is bad for you."	"What else was true that day?" (the book's own question, p. 72)
"I didn't have anything salty, so my page is empty."	"Good — you left it empty because it was empty. That's the whole job."
"Can we test it in class?"	Not with food or drinks. We test on paper here; at home, it's the family's choice, one week only.
"My head hurts."	"Thank you for telling me." Follow your school's usual care, and don't guess the reason out loud — a headache can have many.
"My grandpa has to eat less salt."	"Thanks for sharing. Bodies work differently, and doctors help people take care of theirs." Move on gently.

Family letter

Copy onto school letterhead, or print the one-page version from the hub.

Dear families,

This week our class read The Bottomless Popcorn Paradox, a story about a girl who learns to hear her body's signals — and to check her own reasons — like a scientist.

Your child may bring home a one-week Salty-Snack Experiment. It is optional and private: nothing is collected, checked or graded. It asks your child to notice when they have something salty, how many times they drink water, and how their mouth and head feel — and it treats a blank day and a question mark as real results.

If you'd like to help: ask, don't check. Praise the blank day. If your child names a reason that sounds wrong, let it stand for a minute and ask one question. And stop after one week.

The book never calls a food good or bad and includes no sodium or calorie counts; we're keeping it that way. If you have questions about your child's health, please talk with your pediatrician.

More for families: lumisnutritionlab.com

Corrections to the Lab Notes and Word Lab

Ten passages in Lumi's Lab Notes and Word Lab simplify too far, or say something the evidence doesn't support. Here is what each one says in the book, and what it should say. The rest stands.

SYMPTOM (page 62) — in the book: A clue your body shows on the outside about something happening on the inside.

It should say: A clue you feel on the inside that something is going on in your body. Only you can feel it.

That's how doctors use the word; a clue other people can see, like a rash or a fever, is called a sign. Sources 28, 29.

ATTENTION (page 63) — in the book: …you have zero left for noticing what's happening in your own body.

It should say: …you have less left for noticing what's happening in your own body.

Busy attention turns body feelings down, not off, in studies of adults and of children. Sources 15, 16, 17.

DEHYDRATION (Lumi's note) (page 61) — in the book: Jiwoo didn't notice until too late.

It should say: Jiwoo didn't notice for a long time.

Thirst usually comes before the body is short of water, as page 74 says. Source 8.

EXPERIMENT (Lumi's note) (page 56) — in the book: The movie was the experiment I didn't design but definitely observed.

It should say: The movie wasn't an experiment — nobody changed one thing on purpose. I just observed.

An experiment changes one thing on purpose; watching is observing. Sources 30, 15.

Lab Notes 1 (graph) (page 41) — in the book: SALT IN MY BLOOD · "This is the headache, right on time."

It should say: SALT I ATE · "This is where my head started to hurt."

Blood always has salt, and the body keeps it nearly steady; headaches don't come on a schedule. Sources 1, 2, 11.

Lab Notes 2 (page 42) — in the book: A tight, thirsty cell is what a tight-hat head is made of.

It should say: Special cells in your brain notice the change and ring the thirst bell. Why do some heads ache when they need water? Scientists are still finding out.

How going without water causes headaches is not yet understood. Sources 1, 11.

Lab Notes 3 (page 43) — in the book: It rings as a dry mouth. A tight hat. An ache behind your eyes.

It should say: You feel it as wanting a drink, often with a dry mouth. Jiwoo's head ached too, and a headache can have lots of causes.

Thirst is the urge to drink; a headache is a separate clue that only some people get. Sources 1, 11, 12, 14.

Lab Notes 4 (page 44) — in the book: Either way, that leaves exactly zero for your own dry mouth.

It should say: Either way, there's less left over for your own dry mouth.

Same as ATTENTION. Sources 15, 16, 17.

Lab Notes 6 (page 47) — in the book: Four were true. Only one was the reason.

It should say: Four were true. They didn't all matter the same.

Everything you drink since your last bathroom trip, and how long you wait, adds up. Sources 1, 22.

Lab Notes 7 (page 48) — in the book: A quiet bell can be practiced.

It should say: Listening for a quiet bell can be practiced.

Practice may help you notice; nothing shows it changes the signal itself. Sources 20, 21.

References

Every source behind the science in this book — what each one supports, and who was studied. Most of the research is in adults; we say so under each one. Nothing here is medical advice.

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   A free textbook. Supports how cells lose water to saltier fluid around them, how the hypothalamus senses salty blood and switches on thirst and water-saving, the dry mouth that comes with thirst, and how the bladder stores urine.
2. Suckling RJ, He FJ, Markandu ND, MacGregor GA. Dietary salt influences postprandial plasma sodium concentration and systolic blood pressure. Kidney Int. 2012;81(4):407-411. doi:10.1038/ki.2011.369. doi.org/10.1038/ki.2011.369
   Supports "a whole bucket of popcorn tips the mix": one salty meal raised blood sodium a little. Study in 10 adults.
3. Sodium in your diet. US Food and Drug Administration. Content current as of March 5, 2024. Accessed November 4, 2026. www.fda.gov/food/nutrition-education-resources-materials/sodium-your-diet
   Supports what salt and sodium are, that muscles and nerves need a little, that most of our salt is already in packaged and prepared food, and that children's limits are lower than adults'.
4. About sodium and health. Centers for Disease Control and Prevention. Reviewed June 29, 2026. Accessed November 4, 2026. www.cdc.gov/salt/about/index.html
   Supports the points that the body needs a small amount of sodium, that it turns up in almost every kind of food, and that savory snacks and pizza are among children's top sources.
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   Supports "you need a little every day": sets the small daily amount of sodium children need.
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   Supports the facts that sodium is essential and that too much over time raises blood pressure. International guidance that covers children too.
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